12th IEEE International Conference on Sustainable Technologies and Engineering (IEEE iCOSTE 2026)
1-3 November 2026, Cox's Bazar, Bangladesh

Explainable Multimodal Skin Lesion Classification via Supervised Contrastive Learning and Attention-Based Fusion

Md. Dulal Hossain[1,3,4], Adrita Rahman[4], Maria Akther[1], Umme Ruman[1], Md. Ahsan Habib[2]
[1]Department of Computer Science and Engineering, Green University of Bangladesh, Dhaka, Bangladesh
[2]Department of CSE, Bangladesh University of Business and Technology (BUBT), Dhaka-1216, Bangladesh
[3]Department of Computer Science and Engineering, Jagannath University, 9-10 Chittaranjan Ave, Dhaka 1100
[4]Department of Computer Science and Engineering, Canadian University of Bangladesh (CUB), Dhaka-1212, Bangladesh
Email: dulal.md.cse@gmail.com, adrita.adi141@gmail.com, maria.2021.cse@gmail.com,
ruman@cse.green.edu.bd, ahsan.habib.tareq@gmail.com

***Abstract*—Early skin-lesion diagnosis is challenging due to high visual similarity between lesion categories and severe class imbalance in clinical datasets. We propose *FusionClassifier*, a multimodal framework that combines a Swin-Small Transformer with supervised contrastive learning and attention-based feature fusion. A two-phase training strategy first learns discriminative image representations and then integrates them with patient metadata, including age, sex, and anatomical site. Grad-CAM and Grad-CAM++ are further employed to provide visual explanations of model predictions. Experiments on the HAM10015 dataset using five-fold cross-validation achieve 98.17% accuracy, 97.28% F1-score, and 99.72% AUC-ROC. Ablation results show that supervised contrastive learning substantially improves the Swin-Small baseline from 73.58% to 98.21% accuracy. The results demonstrate the effectiveness of combining discriminative representation learning, multimodal fusion, and explainability for skin-lesion classification.**

***Keywords*—Supervised Contrastive Learning, Skin Cancer Detection, Swin Transformer, Multimodal Fusion, Attention Mechanism, Explainable AI, Deep Learning, Dermatology**

I. Introduction

Skin-lesion classification from dermatoscopic images has gained considerable attention as deep learning enables automated analysis of dermatological conditions [1]. The HAM10015 dataset is a widely used benchmark, containing 10,015 dermatoscopic images across seven diagnostic categories with substantial class imbalance [2]. This imbalance makes minority-class recognition particularly challenging.

Transformer-based architectures have demonstrated strong performance in visual recognition, with Vision Transformers providing effective global image representations [3]. The Swin Transformer further improves efficiency through hierarchical representations and shifted-window self-attention [4]. However, conventional supervised learning may not explicitly encourage discriminative representations between visually similar lesion categories.

Supervised contrastive learning addresses this limitation by bringing samples from the same class closer while separating different classes in the embedding space [5]. It has also shown potential for improving skin-lesion representation learning [6]. In addition, clinical metadata such as age, sex, and anatomical site can provide complementary information beyond image features [7].

Multimodal learning can combine these complementary sources, but simple feature concatenation does not account for their varying importance across individual cases [8]. Attention-based fusion can instead learn modality-specific weights and emphasize the more informative modality [9].

To address these limitations, this paper proposes *FusionClassifier*, which combines a Swin-Small image encoder, supervised contrastive learning, and attention-based multimodal fusion. The first stage learns discriminative image representations, while the second integrates image features with clinical metadata through adaptive modality weighting. Grad-CAM and Grad-CAM++ are further employed to visualize regions contributing to model predictions [10].

On the imbalanced HAM10015 dataset, FusionClassifier achieves 98.17

The main contributions of this work are:

- A supervised contrastive learning strategy with a Swin-Small Transformer for discriminative skin-lesion representation learning.
- An attention-based multimodal fusion mechanism integrating dermatoscopic images with clinical metadata.
- An explainability framework using Grad-CAM and Grad-CAM++.
- Comprehensive evaluation through five-fold stratified group cross-validation and ablation experiments.

The remainder of this paper is organized as follows. Section II reviews related studies on skin lesion classification, multimodal learning, supervised contrastive learning, and explainable artificial intelligence, while also identifying the key research gaps addressed in this study. Section III presents the proposed FusionClassifier framework, including the Swin-Small image encoder, supervised contrastive pretraining, clinical metadata representation, and attention-based multimodal fusion strategy. Section IV describes the datasets, preprocessing procedures, training configuration, evaluation metrics, cross-validation strategy, and experimental results, including per-class analysis and ablation studies. Section V discusses

979-8-3195-2804-9/26/$31.00 ©2026 IEEE

the obtained findings, the contribution of the proposed components, interpretability results, comparative performance, and the limitations of the study. Finally, Section VI summarizes the main contributions and findings of the proposed approach and outlines potential directions for future research, including broader external validation and improved multimodal clinical integration.

II. Literature Review

Automated skin-lesion classification remains challenging due to the visual similarity between different lesion categories and the substantial class imbalance found in widely used dermoscopic datasets. The HAM10015 dataset, for example, contains 10,015 dermatoscopic images distributed across seven diagnostic categories, with considerable differences in the number of samples per class [2]. Such imbalance can make reliable recognition of underrepresented lesions difficult, while visually similar categories can remain challenging to distinguish. The research identifies melanocytic nevi and melanoma as examples of visually similar classes for which explicit representation separation is desirable [11] [12]. Early deep learning approaches demonstrated the potential of automated skin-cancer classification, with Esteva et al. demonstrating dermatologist-level classification performance using deep neural networks [1].

The use of transformer-based architectures has provided an alternative to conventional convolutional feature extraction. Dosovitskiy et al. introduced the Vision Transformer (ViT), demonstrating the effectiveness of transformer architectures for image recognition [3]. The Swin Transformer subsequently introduced a hierarchical architecture with shifted-window self-attention, enabling efficient multi-scale visual representation learning [4]. Despite their strong representation capability, transformer-based classifiers trained with standard classification objectives may not explicitly encourage samples from the same lesion category to form compact representations while separating visually similar classes. This limitation motivates the use of contrastive objectives for learning more discriminative feature spaces.

Another limitation of image-only classification is that dermatoscopic images do not contain all clinically relevant information. Patient metadata, including demographic characteristics and lesion location, can provide complementary contextual information. Adebiyi et al. investigated multimodal learning using HAM10015 [7], while Ou et al. proposed a multimodal framework combining clinical images and metadata for skin-lesion diagnosis [8]. Aboulmira et al. further investigated attention-guided multimodal fusion for skin disease classification [13]. However, existing approaches may rely on direct feature concatenation or other fusion strategies that do not explicitly model the varying importance of individual modalities. The research identifies this as a limitation of existing multimodal approaches and motivates an attention-based fusion mechanism. Attention mechanisms provide a way to dynamically weight information from different sources [9].

Finally, interpretability remains an important consideration for medical AI systems. Grad-CAM provides gradient-based visual explanations by highlighting image regions that contribute to a model's prediction [10], while Grad-CAM++ extends this approach to provide more refined localization [14]. The use of explainability methods has also been explored in multimodal skin-lesion systems, including the XRAI-based framework proposed by Aksoy et al. [15]. However, the research notes that existing approaches generally focus on individual visual explanation methods, while the combined use of Grad-CAM and Grad-CAM++ remains less explored.

TABLE I: Comprehensive Comparison of Skin Lesion Classification Systems

Author(s)	Architecture	Key Innovation	Limitation
Wang et al. [6]	Adversarial + Contrastive.	Modality-invariant representation	Limited to 7-point criteria; no direct disease classification
Adebiyi et al. [7]	ALBEF (AlignBeforeFuse)	Cross-modal alignment prefusion.	Minimal metadata (age, sex, location only)
Wicaksana et al. [16]	YOLOv11 with augmentation	Efficient single-stage detection	No multimodal fusion; limited baseline comparison
Christopoulos [12]	ViT + TRACE (Nested Contrastive)	Pseudo-metadata retrieval	Noisy pseudometadata; poor minority class performance
Aboulmira et al. [13]	EfficientNet + Dense + DS Fusion	Attention-guided belief masses	High computational cost; complex probabilistic fusion
Aksoy et al. [15]	EfficientNet-B3 + XRAI	Clinical deployment platform	Severe dataset imbalance; random split data leakage risk
Kalaiyarivu [17]	CNN + SVM Hybrid	Deep features + traditional classifier	Generic CNN backbone; no metadata; no explainability
Ou et al. [8]	ResNet50 + MLP + Fusion Module	Smartphone compatible multimodal	Small dataset (2,298 images); 6 classes only
Hsu et al. [18]	Dual encoder + HAC Loss	Hierarchy-aware contrastive objective	Complex dualencoder; higher computational overhead
Fusion Classifier (Proposed)	Swin-T (Small) + SupCon + Attention Fusion	Supervised contrastive pretraining + Attention-based multimodal fusion + Dual XAI	Requires complete, high-quality meta-data

Overall, existing research has addressed in Tab. II skin-lesion classification through deep learning, transformer architectures, contrastive learning, multimodal fusion, and explainability. However, these components are often investigated separately. The research identifies gaps in discriminative feature learning, multimodal fusion, the use of Swin Transformer as a multimodal backbone, and comprehensive explainability. This motivates the proposed FusionClassifier, which combines

a Swin Transformer-based image encoder with supervised contrastive learning and attention-based multimodal fusion, with Grad-CAM and Grad-CAM++ employed for visual interpretation.

III. Proposed Methodology

The proposed FusionClassifier is a two-phase multimodal deep learning framework for seven-class skin-lesion classification using dermatoscopic images and clinical metadata. The framework first learns discriminative visual representations through supervised contrastive pretraining and subsequently integrates the learned image features with clinical information using attention-based fusion. The overall workflow includes data collection, preprocessing and augmentation, contrastive representation learning, multimodal feature fusion, classification, and explainability analysis. The two-phase training strategy is illustrated in Fig. 1.

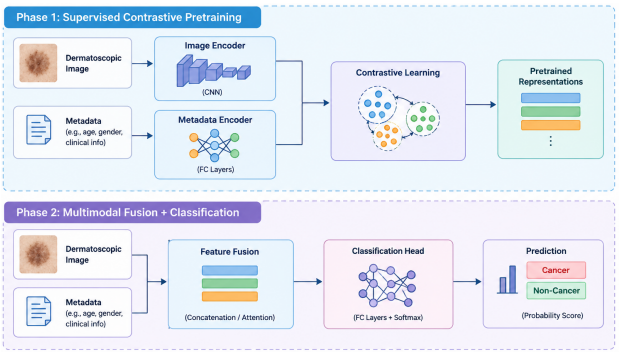


Fig. 1: The proposed two-phase training strategy. Phase 1 focuses on supervised contrastive pretraining, and Phase 2 combines multimodal fusion with classification.

A. System Architecture

The proposed FusionClassifier consists of four main components: a Swin-Small image encoder, clinical metadata encoder, attention-based fusion module, and classification head, as shown in Fig. 2. The framework follows a two-phase training strategy, illustrated in Fig. 1, where visual representation learning is performed before multimodal classification. In Phase 1, the Swin-Small encoder processes the 224 × 224 dermatoscopic images and extracts hierarchical visual features. The resulting 768-dimensional representation is projected into a 128-dimensional embedding space and optimized using supervised contrastive learning, encouraging samples from the same diagnostic class to form similar representations while separating different classes.

In Phase 2, the pretrained image encoder produces a 128-dimensional image feature, while the clinical metadata are transformed into a 128-dimensional representation using a multilayer perceptron. An attention-based fusion module learns the relative importance of the two modalities rather than directly concatenating them. The weighted representations are combined and normalized before being passed to the classification head, which generates predictions for the seven lesion categories. This architecture enables the model to jointly utilize visual characteristics and complementary clinical information for skin-lesion classification.

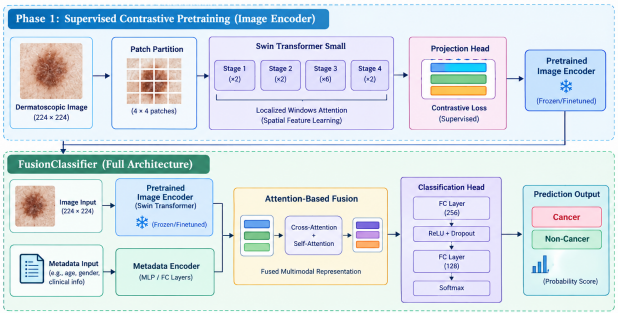


Fig. 2: Architecture of FusionClassifier showing four key components: Swin Transformer encoder pretrained with supervised contrastive learning, metadata processing module, attention-based fusion mechanism, and classification output layer

B. Proposed FusionClassifier

The proposed model consists of two phases. First, the Swin-Small encoder $f_\theta(\cdot)$ extracts image features, which are projected into a 128-dimensional embedding:

$$z_i = g_\psi(f_\theta(x_i)). \tag{1}$$

Supervised contrastive learning is applied using:

$$\mathcal{L}_{SupCon} = -\sum_i \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i^\top z_p/\tau)}{\sum_{a \in A(i)} \exp(z_i^\top z_a/\tau)}, \tag{2}$$

where $\tau = 0.07$, $P(i)$ denotes positive samples, and $A(i)$ denotes the candidate samples.

Clinical metadata is encoded as:

$$m_i = h_\phi(q_i), \tag{3}$$

where q_i contains age, sex, and anatomical site.

The image and metadata features are concatenated:

$$r_i = [z_i; m_i]. \tag{4}$$

Attention weights are obtained by:

$$[\alpha_i^I, \alpha_i^M] = \text{softmax}(W_a r_i + b_a), \tag{5}$$

and the multimodal representation is:

$$f_i = \alpha_i^I z_i + \alpha_i^M m_i. \tag{6}$$

Finally, the seven-class prediction is computed as:

$$\hat{y}_i = \text{Softmax}(W_c f_i + b_c), \tag{7}$$

using the class-weighted cross-entropy loss:

$$\mathcal{L}_{CE} = -\sum_{c=1}^{7} w_c y_{ic} \log(\hat{y}_{ic}). \tag{8}$$

C. Dataset and Data Collection

Experiments are conducted on HAM10015, containing 10,000 dermatoscopic images across seven diagnostic categories: melanocytic nevi (nv), melanoma (mel), benign keratosis (bkl), basal cell carcinoma (bcc), actinic keratoses (akiec), vascular lesions (vasc), and dermatofibroma (df). The dataset is substantially imbalanced, with 6,705 nv images (67.05%) but only 100 df images (1.00%). Associated clinical metadata, including patient age, sex, anatomical site, and diagnosis-related information, are used as the second modality. The complete class distribution is presented in Tab. II.

TABLE II: Comprehensive Comparison of Skin Lesion Classification Systems

Diagnostic Category	**Sample Count**	**Percentage(%)**
Melanocytic nevi (nv)	6,705	66.95%
Melanoma (mel)	1,113	11.11%
Benign keratosis (bkl)	1,099	10.97%
Basal cell carcinoma (bcc)	514	5.13%
Actinic keratoses (akiec)	327	3.27%
Vascular lesions (vasc)	142	1.42%
Dermatofibroma (df)	115	1.15%
Total	10,015	100%

D. Preprocessing

Images are resized to 224 × 224 pixels and normalized using ImageNet statistics. Training augmentation includes random resized cropping, horizontal and vertical flipping, shift-scale-rotation, blur, brightness/contrast adjustment, and color jitter. Two augmented views of each image are generated for supervised contrastive learning. Clinical metadata are processed separately before multimodal fusion. A 5-fold Stratified Group K-Fold strategy is used for evaluation, preserving class distribution while keeping samples associated with the same lesion identifier within a single fold.

E. Proposed Model and Framework

The first phase uses Swin-Small to extract hierarchical visual features from the 224 × 224 input. The encoder produces a 768-dimensional representation, which is projected to a 128-dimensional embedding. For each minibatch, augmented views from the same class form positive pairs, while samples from different classes act as negatives. The supervised contrastive objective encourages intra-class compactness and inter-class separation, using a temperature of 0.07. The pretrained image encoder is then retained for Phase 2. In the second phase, the image encoder generates a 128-dimensional image feature, while a multilayer perceptron transforms the clinical metadata into a 128-dimensional metadata feature. Rather than directly concatenating the two modalities, an MLP with softmax generates attention weights that determine their relative contribution. The weighted features are combined and layer-normalized before being passed to the seven-class classifier. This allows the model to dynamically balance visual and clinical information for each sample.

F. Algorithms and Training

Phase 1 optimizes the Swin-Small encoder and projection head using supervised contrastive loss and Adam optimization. Phase 2 trains the multimodal classification framework using class-weighted cross-entropy, with weights based on inverse class frequency to mitigate the severe class imbalance in HAM10015. The final evaluation uses stratified group 5-fold cross-validation and reports accuracy, precision, recall, specificity, F1-score, and macro-averaged ROC-AUC. Grad-CAM and Grad-CAM++ are subsequently used to visualize image regions contributing to model predictions.

IV. Experimental Setup and Results

A. Experimental Environment

The proposed FusionClassifier was evaluated on the HAM10015 dataset using 5-fold Stratified Group K-Fold cross-validation. The framework was implemented using PyTorch and trained on an NVIDIA Tesla P100 GPU through Kaggle. Training was performed in two phases: supervised contrastive pretraining of the Swin-Small image encoder for 80 epochs, followed by multimodal training for 50 epochs per fold. The final multimodal model employed the Adam optimizer with a learning rate of 10^{-4}, while class-weighted cross-entropy was used to reduce the effect of class imbalance.

B. Evaluation Metrics

Performance was assessed using accuracy, precision, recall, F1-score, specificity, and ROC-AUC. Accuracy measures the overall proportion of correctly classified samples, whereas precision and recall evaluate the reliability and sensitivity of predictions. F1-score provides a balance between precision and recall, specificity measures the correct identification of non-target samples, and ROC-AUC evaluates class discrimination across different decision thresholds. These metrics provide a comprehensive assessment for the imbalanced seven-class skin-lesion classification task.

C. Results

The proposed FusionClassifier achieved a reported 98.17% accuracy, 97.23% precision, 97.40% recall, 97.28% F1-score, 99.58% specificity, and 99.80% ROC-AUC. The fold-wise results demonstrate stable performance, with accuracy values remaining between 98.05% and 98.28% and AUC values between 99.67% and 99.75%. The per-class evaluation also showed strong performance across all seven categories, with F1-scores ranging from 95.32% for melanoma to 99.12% for melanocytic nevi.

TABLE III: Cross-validation results across five folds

Fold	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**AUC**	**Specificity**
1	98.21%	97.56%	96.27%	96.85%	99.75%	99.57%
2	98.05%	97.20%	98.07%	97.63%	99.72%	99.58%
3	98.16%	97.21%	96.61%	96.88%	99.67%	99.53%
4	98.16%	97.75%	98.23%	97.99%	99.73%	99.60%
5	98.28%	96.41%	97.80%	97.05%	99.73%	99.65%
Mean	98.17%	97.23%	97.40%	97.28%	99.72%	99.58%
Std	±0.09%	±0.55%	±0.78%	±0.49%	±0.03%	±0.04%

D. Per-Class Performance Analysis

Tab. III presents the average per-class performance across the five cross-validation folds. The F1-score ranges from 95.32% for melanoma (mel) to 99.12% for melanocytic nevi (nv). Dermatofibroma (df), despite being the least represented class with 100 samples, achieves a 97.41% F1-score and 99.09% recall. Specificity remains high across all classes, ranging from 98.30% to 100%, indicating limited false-positive classification.

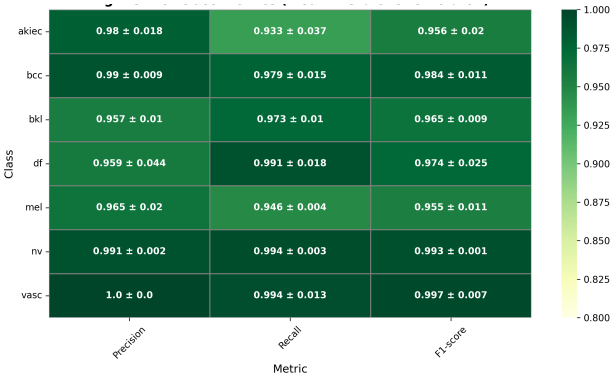


Fig. 3: Per-class performance heatmap across five-fold cross-validation.

Fig. 3 summarizes the per-class performance, showing consistently high precision, recall, and F1-scores across the seven lesion categories.

E. Confusion Matrix Analysis

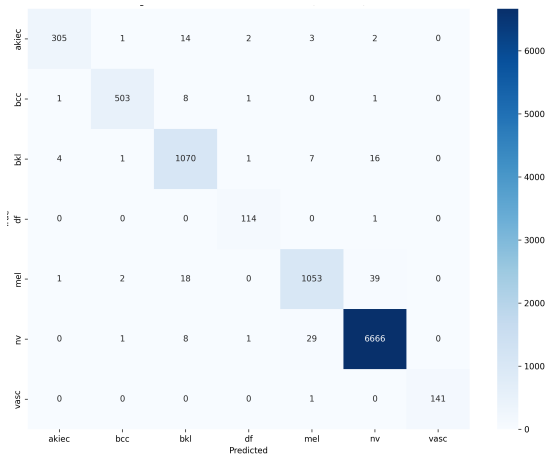


Fig. 4: Average confusion matrix across five-fold cross-validation.

Fig. 4 presents the average confusion matrix across the five cross-validation folds. The strong diagonal dominance indicates that most samples are correctly classified. The main confusion occurs between melanocytic nevi (nv) and benign keratosis (bkl), which have visually similar characteristics. Despite their limited samples, the minority classes, particularly dermatofibroma (df) and vascular lesions (vasc), show relatively low misclassification.

F. Comparative Analysis

Compared with the approaches reported in the research, FusionClassifier provides competitive overall performance. As shown in Fig. 5. It achieves 98.17% accuracy, 97.28% F1-score, and 99.80% AUC, compared with 95.40% accuracy, 94.80% F1-score, and 98.50% AUC reported by Wang et al. [6], and 97.10% accuracy, 96.50% F1-score, and 99.10% AUC reported by Adebiyi et al. [7]. The improvement demonstrates the benefit of combining discriminative visual representation learning with complementary clinical metadata through multimodal fusion.

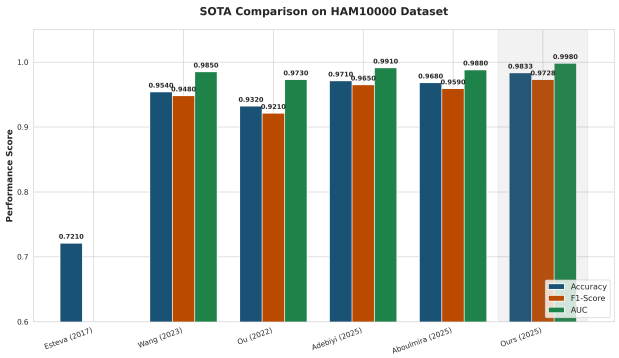


Fig. 5: Performance benchmarking against comparison with existing methods on HAM10015

G. Ablation and Explainability

The ablation results indicate that supervised contrastive pretraining is the primary contributor to the improvement in classification performance in Tab. IV. The Swin-Small baseline improves from 73.58% to 98.21% accuracy after supervised contrastive pretraining. The subsequent multimodal experiments show that incorporating clinical metadata provides complementary information, while attention-based fusion achieves a slightly higher performance than direct feature concatenation under the same cross-entropy objective. These results suggest that the principal gain originates from discriminative visual representation learning, whereas attention-based multimodal fusion provides an additional, albeit smaller, refinement by adaptively weighting image and metadata features.

TABLE IV: Ablation study: impact of model components

Configuration	Accuracy	Precision	Recall	F1-Score	AUC
Swin-Small (baseline)	0.735	0.562	0.721	0.609	0.946
+ SupCon Pretraining	0.982	0.973	0.973	0.973	0.996
+ Metadata (Concat, FL)	0.975	0.969	0.885	0.915	0.997
+ Metadata (Attn, FL)	0.982	0.967	0.973	0.970	0.997
+ Metadata (Concat, CE)	0.982	0.973	0.972	0.972	0.997
Full Model (Attn, CE)	0.983	0.972	0.973	0.972	0.997

For qualitative interpretability, Grad-CAM and Grad-CAM++ were applied to visualize the image regions contributing to model predictions. As shown in Fig. 6, the activation maps provide visual evidence of the regions emphasized by the model across the seven lesion categories, supporting the interpretability of the classification decisions [10] [14].

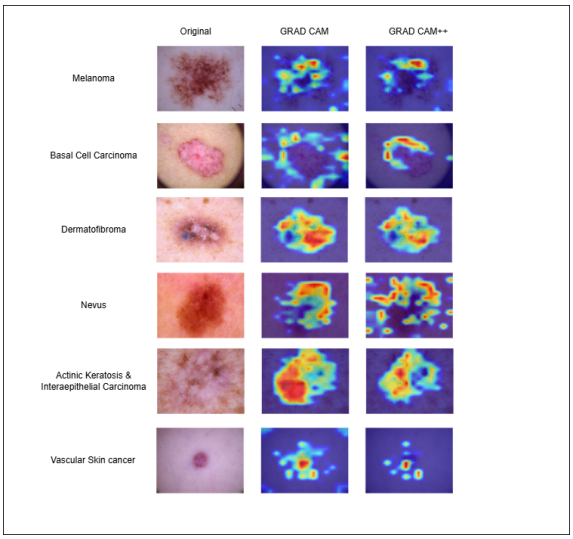


Fig. 6: Visualization of seven classes usinf GRAD-CAM and GRAD-CAM++

V. Discussion

The results show that FusionClassifier effectively addresses visual similarity, class imbalance, multimodal information, and interpretability in skin lesion classification. The model achieved 98.17% accuracy, 97.28% F1-score, and 99.72% ROC-AUC, with consistent fold-wise performance. Supervised contrastive pretraining substantially improved the Swin-Small baseline from 73.58% to 98.21% accuracy, demonstrating its effectiveness in learning discriminative lesion representations.

The multimodal experiments further show that attention-based fusion provides effective integration of image and clinical features compared with direct concatenation. This highlights the complementary value of visual representations and clinical metadata. Grad-CAM and Grad-CAM++ provided qualitative evidence of model interpretability by highlighting image regions contributing to predictions. However, these visualizations should not be considered complete clinical explanations. Further external validation is required to assess generalizability beyond HAM10015.

VI. Conclusion and Future Work

This study proposed FusionClassifier, a multimodal framework combining a Swin-Small encoder, supervised contrastive pretraining, and attention-based fusion of dermatoscopic images and clinical metadata. On HAM10015, the model achieved 98.17% accuracy, 97.28% F1-score, and 99.72% ROC-AUC, while ablation results confirmed the importance of supervised contrastive learning and attention-based fusion. Grad-CAM and Grad-CAM++ further provided qualitative interpretability.

Future work will focus on external validation using diverse datasets, incorporation of additional clinical information, improved multimodal fusion strategies, and prospective evaluation with dermatology specialists. These efforts can further assess the generalizability and practical usefulness of the proposed framework.

References

[1] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.

[2] P. Tschandl, C. Rosendahl, and H. Kittler, "The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Scientific Data*, vol. 5, p. 180161, 2018.

[3] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020. [Online]. Available: https://arxiv.org/abs/2010.11929

[4] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 10 012–10 022.

[5] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, "Supervised contrastive learning," *arXiv preprint arXiv:2004.11362*, 2020. [Online]. Available: https://arxiv.org/abs/2004.11362

[6] Z. Wang, L. Zhang, X. Shu, Y. Wang, and Y. Feng, "Consistent representation via contrastive learning for skin lesion diagnosis," *Computer Methods and Programs in Biomedicine*, 2023.

[7] A. Adebiyi, N. Abdalnabi, E. H. Smith, J. Hirner, E. J. Simoes, M. Becevic, and P. Rao, "Accurate skin lesion classification using multimodal learning on the ham10000 dataset," *medRxiv*, 2024. [Online]. Available: https://www.medrxiv.org/content/early/2024/05/31/2024.05.30.24308213

[8] C. Ou, S. Zhou, R. Yang, W. Jiang, H. He, W. Gan, W. Chen, X. Qin, W. Luo, X. Pi, and J. Li, "A deep learning based multimodal fusion model for skin lesion diagnosis using smartphone collected clinical images and metadata," *Frontiers in Surgery*, vol. 9, p. 1029991, 2022.

[9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," *arXiv preprint arXiv:1706.03762*, 2017. [Online]. Available: https://arxiv.org/abs/1706.03762

[10] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-cam: Visual explanations from deep networks via gradient-based localization," in *2017 IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 618–626.

[11] A. S. Mohammed, M. S. Mohammed Ali, S. M. J. Abdalwahid, and S. W. Kareem, "Contrastive self-supervised ensemble transfer learning for robust skin cancer classification and early detection," *Asian Pacific Journal of Cancer Prevention*, vol. 26, no. 7, pp. 2607–2617, 2025.

[12] D. Christopoulos, S. Spanos, E. Baltzi, V. Ntouskos, and K. Karantzalos, "Skin lesion phenotyping via nested multi-modal contrastive learning," *arXiv preprint arXiv:2505.23709*, 2025. [Online]. Available: https://arxiv.org/abs/2505.23709

[13] A. Aboulmira, H. Hrimech, M. Lachgar, H. Malyana, and K. Lemouden, "Attention-guided multimodal skin disease classification using discounted dempster-shafer fusion and efficientnet," *Informatics in Medicine Unlocked*, vol. 58, p. 101699, 10 2025.

[14] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. Balasubramanian, "Grad-cam++: Generalized gradient-based visual explanations for deep convolutional networks," 10 2017.

[15] S. Aksoy, P. Demircioglu, and I. Bogrekci, "Web-based multimodal deep learning platform with xrai explainability for real-time skin lesion classification and clinical decision support," *Cosmetics*, vol. 12, no. 5, p. 194, 2025.

[16] I. Wicaksana, R. Pramunendar, G. Sarasati, and G. Trisnapradika, "Skin lesion classification using yolov11 on the ham10000 dataset," *Inform : Jurnal Ilmiah Bidang Teknologi Informasi dan Komunikasi*, vol. 10, pp. 45–52, 01 2025.

[17] M. Kalaiyarivu and N. J. Nalini, "Skin disease classification using combined machine learning and deep learning models," *Journal of Pharmaceutical Negative Results*, vol. 13, no. Special Issue 9, pp. 1847–1855, 2022.

[18] B. W.-Y. Hsu and V. S. Tseng, "Hierarchy-aware contrastive learning with late fusion for skin lesion classification," *Computer Methods and Programs in Biomedicine*, vol. 226, p. 106666, 2022.